

Scientists find the earliest known animal

Scientists had found the fossils of an ancient creature 1.4 meters in length, called the Dickinsonia. However, they did not know what kind of an organism it was, a giant unicellular organism, a colony of simpler organisms, lichen, or the earliest known animal ancestor. For over 75 years, researchers did not know the true nature of this creature. A team of scientists studying a well preserved fossil from a remote area in the White Sea found molecules of cholesterol still in the fossil. The fat is a sign of animal life, which finally solved the long standing mystery. The discovery pushes back the current understanding of when animal life became common on Earth, and shows that large animals were prolific even 558 million years ago. The discovery of the fat molecules now



confirms that Dickinsonia is the oldest known animal. Normally, scientists study only the structure of the fossils, and do not look for any molecular traces. However, this team of scientists had the crazy idea of looking for molecules, considering how well preserved the fossil was.

<https://dgit.in/eanim>

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THIS MONTH IN SCIENCE:

From solving the ocean's garbage problem to putting entire labs on a chip, science has a lot to enthrall you this month. Gear up as India plans its first manned space mission before you find out where the CRISPR vs autism debate stands.

The international neutrino experiment prototype

The International Deep Underground Neutrino Experiment (DUNE) has kicked off with the discovery of particle traces in the world's largest liquid argon neutrino detector. The prototype detector is a three storey high cube called the ProtoDUNE, and has been built at CERN. The detector is filled with 800 tons of liquid argon, which keeps the temperatures below -184°C . The detector can be used to track



particles from both cosmic rays and neutrino beams produced by accelerators. The Proto-DUNE is the first of two tests, with the final larger detector, to be built in the USA. The final

DUNE detector will be built on the same lines as ProtoDUNE. The neutrino detector will be used to probe some of the most pressing problems in science, including why matter is much more common than antimatter, and more. The final DUNE detector will study the most intense neutrino beam in the world, and will be located 1,300 kilometers from the source.

<https://dgit.in/dunexp>



Circuits for living cells developed

Researchers from Caltech have developed a toolkit of proteins that can be assembled together like Lego bricks for various purposes. One of the proof of concept implementations was a circuit that was added to human cells to detect cancer. If cancer was detected in the cells, then the circuit would cause the cancerous cells to self destruct.

<https://dgit.in/cellcircuit>



The Sahara can power the planet

Massive solar and wind power farms in the Sahara desert, built using existing tech can generate about 21 terawatt hours of electricity, enough to power the entire world. Additionally, such farms would increase rainfall and stop the desert from growing any further.

<https://dgit.in/saharafarm>



Toddlers can weigh risks

New research has shown that even toddlers perform cost-benefit analyses, similar to how adults consider risks and rewards before taking an action. When asked to move blocks and hand it over, the toddlers would follow the request based on block weight.

<https://dgit.in/babyblocks>



"Ghost Galaxies" discovered

A census of over 11,000 galaxies has detected a population of galaxies that have previously been considered to be unobservable. These ghost galaxies simply do not emit enough light to be detected using conventional methods of astronomy.

<https://dgit.in/ghostgalaxy>